

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

- In a series AC circuit with load resistance R connected across a DC source , the instantaneous load power is $I_d^2 R$ for all x . Under DC operation , the average power dissipated is the same as the average power delivered : $P_{avg} = V_{total} I_d = IR$.
- With $V_{total} = 12 \text{ V}$ and $P_{avg} = 40 \text{ W}$, solve for current : $I = P_{avg}/V_{total} = 40/12 = 3.33 \text{ A}$.
- The required resistance of the load is given by $P = I^2 R$, so $R = P/I = (40/12)/3.33 \hat{=} 4.76 \hat{\odot} \hat{=} 4.8 \hat{\odot}$.

Therefore , the correct answer is : I . 2 . 5 amps , 4 . 8 oh ms

</think>

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- With $V_{\text{total}} = 12 \text{ V}$ and $P_{\text{avg}} = 40 \text{ W}$, solve for current: $I = P_{\text{avg}} / V_{\text{total}} = 40 / 12 = 3.33 \text{ A}$.

- The required resistance of the load is given by $P = I^2 R$, so $R = P/I^2 = (40/12)/3.33 \approx 4.76 \approx 4.8 \Omega$.

Therefore, the correct answer is: 1.2.5 amps, 4.8 ohms

</think>

< answer >

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</answer>